# Confluent Platform with Apache Flink Minikube Deployment

A Makefile-driven quickstart that deploys a full local streaming stack on Minikube:

- **Confluent Platform** (KRaft mode) via Confluent for Kubernetes (CFK)
- **Apache Flink 2.1.1** via the Confluent Flink Kubernetes Operator 1.130
- **Confluent Manager for Apache Flink (CMF) 2.1** for Flink environment management
- **Kafka UI** (Provectus) for cluster inspection

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## 1.0 Prerequisites

macOS with Homebrew. To install all required tools in one step:

```
make install-prereqs
```

This installs Docker Desktop, `kubectl`, and Minikube via Homebrew. Once complete, **launch Docker Desktop** before proceeding.

To verify all tools are present without installing:

```
make check-prereqs
```

Required: `docker`, `kubectl`, `minikube`, `helm`, `envsubst` (`brew install gettext`).

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## 2.0 Resource Requirements

Minikube is configured with the following defaults, which are required to run the full stack:

| Resource | Default |
|---|---|
| CPUs | 6 |
| Memory | 20 GB |
| Disk | 50 GB |

Override any of these at the command line:

```
make cp-up MINIKUBE_CPUS=8 MINIKUBE_MEM=24576
```

---

## 3.0 Architecture

```
graph TD
    %% ── Composite entry points ──────────────────────────────
    CP_UP(["`**make cp-up**`"])
    FLINK_UP(["`**make flink-up**`"])
    CP_DOWN(["`**make cp-down**`"])
    FLINK_DOWN(["`**make flink-down**`"])
    TEARDOWN(["`**make confluent-teardown**`"])

    %% ── Phase 1: Prerequisites ──────────────────────────────
    subgraph P1["Phase 1 — Prerequisites"]
        CHECK_PRE["check-prereqs\ndocker · kubectl · minikube · helm"]
    end

    %% ── Phase 2: Minikube ───────────────────────────────────
    subgraph P2["Phase 2 — Minikube"]
        MK_START["minikube-start\ncpus=6 · mem=20GB · disk=50GB"]
        MK_STOP["minikube-stop"]
    end
```

```
    %% ── Phase 3: Confluent Operator
────────────────────────────────────────
    subgraph P3["Phase 3 — Confluent Operator"]
        NS["namespace\nkubectl create namespace confluent"]
        OP_INSTALL["operator-install\nhelm: confluent-for-kubernetes"]
        OP_UNINSTALL["operator-uninstall"]
    end

    %% ── Phase 4: Confluent Platform
────────────────────────────────────────
    subgraph P4["Phase 4 — Confluent Platform"]
        CP_DEPLOY["cp-deploy\nKafka KRaft · SR · Connect\nksqlDB · REST
Proxy · C3"]
        CP_DELETE["cp-delete"]
    end

    %% ── Phase 5: Control Center
──────────────────────────────────────────
    subgraph P5["Phase 5 — Control Center"]
        C3["c3-open\nlocalhost:9021"]
    end

    %% ── Phase 6: Apache Flink
───────────────────────────────────────────
    subgraph P6["Phase 6 — Apache Flink"]
        CERT["flink-cert-manager\ncert-manager v1.18.2"]
        FL_OP["flink-operator-install\nhelm: confluentinc/flink-
kubernetes-operator 1.130.0"]
        FL_DEPLOY["flink-deploy\nenvsubst → FlinkDeployment CR\ncp-
flink:2.1.1-cp1"]
        FL_UI["flink-ui\nlocalhost:8081"]
        FL_DELETE["flink-delete"]
        FL_OP_UN["flink-operator-uninstall"]
        CERT_UN["cert-manager-uninstall"]
    end

    %% ── Phase 7: CMF
─────────────────────────────────────────────────────
    subgraph P7["Phase 7 — Confluent Manager for Apache Flink (CMF)"]
        CMF_INSTALL["cmf-install\nhelm: confluent-manager-for-apache-flink
2.1.0"]
        CMF_ENV["cmf-env-create\nPOST /cmf/api/v1/environments"]
        CMF_OPEN["cmf-open\nlocalhost:8080/cmf/api/v1/environments"]
        CMF_PROXY["cmf-proxy-inject\nsocat sidecar → C3 Flink tab"]
        CMF_UN["cmf-uninstall"]
    end

    %% ── Phase 8: Kafka UI
──────────────────────────────────────────────
    subgraph P8["Phase 8 — Kafka UI"]
        KUI_INSTALL["kafka-ui-install\nhelm: provectus/kafka-
ui\nbootstrap: kafka:9092"]
        KUI_OPEN["kafka-ui-open\nllocalhost:8080"]
        KUI_UN["kafka-ui-uninstall"]
```

```
    end

    %% ── Manifests / Templates ──────────────────────────────────────────
    subgraph FS["k8s/base/"]
        MANIFEST[("flink-basic-deployment.yaml\nFLINK_IMAGE · FLINK_VERSION")]
    end

    %% ── make cp-up dependency chain ────────────────────────────────────
    CP_UP --> CHECK_PRE
    CP_UP --> MK_START
    CP_UP --> CP_CORE_UP
    CP_UP --> KUI_INSTALL

    CP_CORE_UP["cp-core-up"] --> OP_INSTALL
    CP_CORE_UP --> CP_DEPLOY
    OP_INSTALL --> NS

    %% ── make flink-up dependency chain ─────────────────────────────────
    FLINK_UP --> CERT
    FLINK_UP --> FL_OP
    FLINK_UP --> CMF_INSTALL
    FLINK_UP --> CMF_ENV
    FLINK_UP --> FL_DEPLOY
    FL_OP --> NS
    FL_DEPLOY --> MANIFEST

    %% ── make cp-down dependency chain ──────────────────────────────────
    CP_DOWN --> KUI_UN
    CP_DOWN --> CP_DELETE
    CP_DOWN --> OP_UNINSTALL

    %% ── make flink-down dependency chain ───────────────────────────────
    FLINK_DOWN --> FL_DELETE
    FLINK_DOWN --> CMF_UN
    FLINK_DOWN --> FL_OP_UN
    FLINK_DOWN --> CERT_UN

    %% ── make confluent-teardown ────────────────────────────────────────
    TEARDOWN -->|"1 — check minikube running"| FLINK_DOWN
    TEARDOWN -->|"2"| CP_DOWN
    TEARDOWN -->|"3 — delete namespace"| NS
    TEARDOWN -->|"4"| MK_STOP

    %% ── UI access ──────────────────────────────────────────────────────
    CP_DEPLOY -.->|"once Running"| C3
    FL_DEPLOY -.->|"once Running"| FL_UI
```

```
    CMF_INSTALL -.->|"once Running"| CMF_OPEN
    CMF_INSTALL -.->|"C3 Flink tab"| CMF_PROXY
    KUI_INSTALL -.->|"once Running"| KUI_OPEN

    %% ── Styles
────────────────────────────────────────────────────────────────
    classDef entry    fill:#1a1a2e,stroke:#e94560,color:#fff,font-
weight:bold
    classDef install  fill:#16213e,stroke:#0f3460,color:#a8dadc
    classDef remove   fill:#2d1b1b,stroke:#8b0000,color:#ffb3b3
    classDef ui       fill:#1b2d1b,stroke:#2d6a2d,color:#b3ffb3
    classDef file     fill:#2d2b1b,stroke:#8b7500,color:#ffe680
    classDef composite fill:#2a1a2e,stroke:#9b59b6,color:#dbb8ff

    class CP_UP,FLINK_UP,CP_DOWN,FLINK_DOWN,TEARDOWN entry
    class CP_CORE_UP composite
    class
CHECK_PRE,MK_START,NS,OP_INSTALL,CP_DEPLOY,CERT,FL_OP,FL_DEPLOY,CMF_INSTAL
L,CMF_ENV,CMF_PROXY,KUI_INSTALL install
    class
MK_STOP,OP_UNINSTALL,CP_DELETE,FL_DELETE,FL_OP_UN,CERT_UN,CMF_UN,KUI_UN
remove
    class C3,FL_UI,CMF_OPEN,KUI_OPEN ui
    class MANIFEST file
```

---

## 4.0 Quickstart

### 4.1 Full stack (CP + Kafka UI)

```
make cp-up
```

This runs: `check-prereqs` → `minikube-start` → `namespace` → `operator-install` → `cp-deploy` → `kafka-ui-install`.

Once pods are up, open Control Center:

```
make c3-open        # http://localhost:9021
```

### 4.2 Add Apache Flink + CMF (run separately after `make cp-up`)

```
make flink-up
```

This runs: `flink-cert-manager` → `flink-operator-install` → `cmf-install` → `cmf-env-create` → `flink-deploy`. `flink-up` is self-contained and can also be run standalone on a fresh cluster.

Once the Flink JobManager pod is running:

```
make flink-ui       # http://localhost:8081
make cmf-open       # http://localhost:8080/cmf/api/v1/environments
```

To expose the Flink tab inside Control Center, inject the CMF proxy sidecar:

```
make cmf-proxy-inject
```

---

## 5.0 Composite Workflow Reference

| Target | What it does |
|---|---|
| `make cp-up` | Full stack: Minikube + CP + Kafka UI |
| `make flink-up` | cert-manager + Confluent Flink Operator + CMF + Flink cluster |
| `make cp-down` | Remove CP, Kafka UI, and Operator (Minikube keeps running) |
| `make flink-down` | Remove Flink cluster, CMF, Operator, and cert-manager |
| `make confluent-teardown` | Full teardown: everything + stop Minikube |

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## 6.0 Individual Target Reference

### 6.1 Phase 1 — Prerequisites

| Target | Description |
|---|---|
| `install-prereqs` | Install Docker Desktop, kubectl, Minikube via Homebrew |
| `check-prereqs` | Verify all required tools are available |

### 6.2 Phase 2 — Minikube

| Target | Description |
|---|---|
| `minikube-start` | Start Minikube with configured resources |
| `minikube-status` | Show Minikube and node status |
| `minikube-stop` | Stop the Minikube cluster |
| `minikube-delete` | Permanently delete the Minikube cluster |

### 6.3 Phase 3 — Confluent Operator

| Target | Description |
|---|---|

| Target | Description |
|---|---|
| `namespace` | Create the `confluent` namespace and set it as default context |
| `operator-install` | Add Confluent Helm repo and install CFK Operator |
| `operator-status` | Show CFK Operator pod status |
| `operator-uninstall` | Remove the CFK Operator Helm release |

### 6.4 Phase 4 — Confluent Platform

| Target | Description |
|---|---|
| `cp-deploy` | Deploy Kafka (KRaft), Schema Registry, Connect, ksqlDB, REST Proxy, Control Center |
| `cp-watch` | Watch pod startup live (Ctrl+C to exit) |
| `cp-status` | Show current pod status |
| `cp-delete` | Remove all CP components and leftover PVCs |

### 6.5 Phase 5 — Control Center

| Target | Description |
|---|---|
| `c3-open` | Port-forward Control Center and open `http://localhost:9021` |

### 6.6 Phase 6 — Apache Flink

| Target | Description |
|---|---|
| `flink-cert-manager` | Install cert-manager (Confluent Flink Operator dependency) |
| `flink-operator-install` | Install the Confluent Flink Kubernetes Operator (`confluentinc/flink-kubernetes-operator`) |
| `flink-operator-status` | Show Flink Operator pod status |
| `flink-operator-uninstall` | Remove the Confluent Flink Operator Helm release |
| `flink-deploy` | Deploy the Flink session cluster using `FLINK_MANIFEST` |
| `flink-status` | Show Flink pods and FlinkDeployment CRs |
| `flink-ui` | Port-forward Flink UI and open `http://localhost:8081` |
| `flink-delete` | Delete the Flink session cluster |
| `cert-manager-uninstall` | Remove cert-manager |

### 6.7 Phase 7 — Confluent Manager for Apache Flink (CMF)

| Target | Description |
|---|---|
| `cmf-install` | Install CMF via Helm (`confluent-manager-for-apache-flink`) and wait for pod readiness |
| `cmf-env-create` | Create a Flink environment (`CMF_ENV_NAME`) in CMF pointing to the `confluent` namespace |
| `cmf-status` | Show CMF pod status and list registered Flink environments |
| `cmf-open` | Port-forward CMF REST API and open `http://localhost:8080/cmf/api/v1/environments` |
| `cmf-uninstall` | Uninstall CMF (safe to run even if not installed) |
| `cmf-proxy-inject` | Patch the C3 StatefulSet with a `socat` sidecar to expose the Flink tab in Control Center |
| `cmf-proxy-remove` | Remove the CMF proxy sidecar and resume CFK reconciliation |
| `cmf-proxy-logs` | Stream logs from the `cmf-proxy` sidecar in the C3 pod |

### 6.8 Phase 8 — Kafka UI (Provectus)

| Target | Description |
|---|---|
| `kafka-ui-install` | Install Kafka UI connected to the local CP cluster (Kafka + Schema Registry + Connect) |
| `kafka-ui-status` | Show Kafka UI pod status |
| `kafka-ui-open` | Port-forward Kafka UI and open `http://localhost:8080` |
| `kafka-ui-uninstall` | Remove Kafka UI |

## 7.0 Configuration

All variables are overridable at the command line. Defaults:

| Variable | Default | Description |
|---|---|---|
| `NAMESPACE` | `confluent` | Kubernetes namespace |
| `CONFLUENT_MANIFEST` | `k8s/base/confluent-platform-c3++.yaml` | Path to Confluent Platform manifest |
| `MINIKUBE_CPUS` | `6` | vCPUs allocated to Minikube |
| `MINIKUBE_MEM` | `20480` | Memory in MB |
| `MINIKUBE_DISK` | `50g` | Disk size |

| Variable | Default | Description |
|---|---|---|
| `FLINK_OPERATOR_VER` | `1.130.0` | Confluent Flink Kubernetes Operator version |
| `FLINK_IMAGE` | `confluentinc/cp-flink:2.1.1-cp1-java21-arm64` | Flink container image |
| `FLINK_VERSION` | `v2_1` | Flink API version string for the FlinkDeployment CR |
| `FLINK_CLUSTER_NAME` | `flink-basic` | Name of the FlinkDeployment resource |
| `FLINK_MANIFEST` | `k8s/base/flink-basic-deployment.yaml` | Path to FlinkDeployment template |
| `CERT_MANAGER_VER` | `v1.18.2` | cert-manager version |
| `CMF_VER` | `2.1.0` | Confluent Manager for Apache Flink version |
| `CMF_PORT` | `8080` | CMF REST API local port |
| `CMF_ENV_NAME` | `dev-local` | Flink environment name registered in CMF |
| `C3_PORT` | `9021` | Control Center local port |
| `FLINK_UI_PORT` | `8081` | Flink UI local port |
| `KAFKA_UI_PORT` | `8080` | Kafka UI local port |

> **Note:** CMF uses the Confluent-packaged Flink operator (`confluentinc/flink-kubernetes-operator`) and `confluentinc/cp-flink` images — not the Apache OSS Flink operator or `flink` Docker Hub image.

Example — deploy a specific Flink image:

```
make flink-deploy FLINK_IMAGE=confluentinc/cp-flink:2.1.1-cp1-java21
FLINK_VERSION=v2_1
```

---

## 8.0 Repository Layout

```
.
├── Makefile
├── README.md
├── README.pdf
├── CHANGELOG.md
├── CHANGELOG.pdf
```

```
├── KNOWN_ISSUES.md
├── KNOWN_ISSUES.pdf
├── LICENSE.md
├── LICENSE.pdf
├── .gitignore
├── docs
│   ├── manual_deployment.md            # Step-by-step manual deployment
instructions (without Makefile)
│   └── manual_deployment.pdf
└── k8s/
    └── base/
        ├── confluent-platform-c3++.yaml    # Confluent Platform manifest
(KRaft + all components)
        └── flink-basic-deployment.yaml     # FlinkDeployment CR template
```

> The `flink-basic-deployment.yaml` is a template — `FLINK_IMAGE` and `FLINK_VERSION` are substituted at deploy time via `envsubst`. Do not apply it directly with `kubectl apply`.

---

## 9.0 Teardown

Remove everything and stop Minikube:

```
make confluent-teardown
```

To keep Minikube running but remove all deployed components:

```
make flink-down   # Flink cluster + CMF + operator + cert-manager
make cp-down      # CP + Kafka UI + CFK Operator
```

---

## 10.0 Manual Deployment Instructions

For users who want to understand the underlying steps without using the Makefile, see docs/manual_deployment.md.

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## 11.0 Resources

- Manage Confluent Platform with Confluent for Kubernetes
- Get Started with Confluent Platform for Apache Flink